

Q: What is the primary function of Matplotlib in machine learning?

A: The primary function of Matplotlib is to visualize data using graphs and charts.

QUESTION 1 Student Answer: ``python "The primary function of Matplotlib is to visualize data using graphs and charts."`` Correct Answer: Matplotlib is a visualization library in Python used in machine learning to present data in charts, lines, bars, and other graphical formats. Score: 6 Feedback: Missing explanation. Matplotlib is used to visualize data, but the student didn't provide a clear explanation of its primary function.

Q: What is the main difference between supervised and unsupervised learning in machine learning?

A: Supervised learning uses labeled data

QUESTION 2 Student Answer: ``python "Supervised learning uses labeled data"`` Correct Answer: Supervised learning uses labeled data, while unsupervised learning uses unlabeled data. Score: 3 Feedback: Missing explanation. The student only mentioned that supervised learning uses labeled data, but didn't explain the difference between supervised and unsupervised learning.

Q: What is the purpose of Data Cleaning in the data science process?

A: Data cleaning removes errors and improves the quality of data.

QUESTION 3 Student Answer: ``python "Data cleaning removes errors and improves the quality of data."`` Correct Answer: Data Cleaning is the process of detecting and removing errors and inconsistencies from input data. It improves data quality and is commonly done using Pandas. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the purpose of data cleaning in the context of data science.

Q: What is the role of Pandas in data science?

A: Pandas is used for loading, cleaning, and manipulating data.

QUESTION 4 Student Answer: ``python "Pandas is used for loading, cleaning, and manipulating data."`` Correct Answer: Pandas is a Python library used in data science for loading data from sources like Excel and CSV files and for cleaning and manipulating data. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the role of Pandas in data science.

Q: What is the difference between fillna() and dropna() functions in Pandas?

A: . fillna() fills missing values, while dropna() removes missing data.

QUESTION 5 Student Answer: ``python ".fillna() fills missing values, while dropna() removes missing data."`` Correct Answer: fillna() replaces missing values, while dropna() removes rows or columns with missing values. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the difference between fillna() and dropna().

Q: What is Imputing in the context of data science, and what strategies are used to fill missing values?

A: Imputing fills missing values using strategies like mean, median, or mode.

QUESTION 6 Student Answer: ``python "Imputing fills missing values using strategies like mean, median, or mode."`` Correct Answer: Imputing is the process of filling missing values using strategies like mean, median, or mode. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the purpose of imputing in the context of data science.

Q: What is the purpose of the fit() method in machine learning?

A: The fit() method is used to train a machine learning model.

QUESTION 7 Student Answer: ``python "The fit() method is used to train a machine learning model."`` Correct Answer: The fit() method is used to train a machine learning model, but the student didn't explain the purpose of the fit() method. Score: 3 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the purpose of the fit() method.

Q: What is the difference between Mean Squared Error (MSE) and accuracy_score() in evaluating regression model performance?

A: MSE measures error in regression, while accuracy_score() measures accuracy in classification.

QUESTION 8 Student Answer: ``python "MSE measures error in regression, while accuracy\_score() measures accuracy in classification."`` Correct Answer: MSE measures error in regression, while accuracy_score() measures accuracy in classification. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the difference between MSE and accuracy_score().

Q: What is the main advantage of using Neural Networks in machine learning?

A: Neural networks can learn complex patterns and improve prediction accuracy.

QUESTION 9 Student Answer: ``python "Neural networks can learn complex patterns and improve prediction accuracy." `` Correct Answer: Neural networks can learn complex patterns and improve prediction accuracy. Score: 9 Feedback: Correct answer. The student provided a clear and concise explanation of the main advantage of using Neural Networks in machine learning.

Q: What is the primary function of a Confusion Matrix in evaluating classification performance?

A: A confusion matrix evaluates classification performance using true/false positives and negatives.

QUESTION 10 Student Answer: ``python "A confusion matrix evaluates classification performance using true/false positives and negatives." `` Correct Answer: A confusion matrix evaluates classification performance using true positive, false positive, true negative, and false negative values. Score: 3 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the purpose of a confusion matrix.

Q: What is the difference between Linear Regression and Support Vector Regression in regression algorithms?

A: Linear Regression models linear relationships, while Support Vector Regression handles non-linear data using kernels.

QUESTION 11 Student Answer: ``python "Linear Regression models linear relationships, while Support Vector Regression handles non-linear data using kernels." `` Correct Answer: Linear Regression models linear relationships, while Support Vector Regression handles non-linear data using kernels. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the difference between Linear Regression and Support Vector Regression.

Q: What is the role of Deep Learning in machine learning?

A: Deep learning uses neural networks to solve complex problems with large datasets.

QUESTION 12 Student Answer: ``python "Deep learning uses neural networks to solve complex problems with large datasets." `` Correct Answer: Deep learning uses neural networks to solve complex problems with large datasets. Score: 9 Feedback: Correct answer. The student provided a clear and concise explanation of the role of Deep Learning in machine learning.

Q: What is the main difference between Random Forest and Decision Tree in classification algorithms?

A: Decision Tree uses a single tree, while Random Forest uses multiple trees for better accuracy.

QUESTION 13 Student Answer: ``python "Decision Tree uses a single tree, while Random Forest uses multiple trees for better accuracy." `` Correct Answer: Decision Tree uses a single tree, while Random Forest uses multiple trees for better accuracy. Score: 6 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the difference between Random Forest and Decision Tree.

Q: What is the primary function of a Convolutional Neural Network (CNN) in machine learning?

A: Linear Regression models linear relationships, while Support Vector Regression handles non-linear data using kernels.

QUESTION 14 Student Answer: ``python "Linear Regression models linear relationships, while Support Vector Regression handles non-linear data using kernels." `` Correct Answer: Linear Regression models linear relationships, while Support Vector Regression handles non-linear data using kernels. Score: 3 Feedback: Missing explanation. The student provided a correct statement, but didn't explain the difference between Linear Regression and Support Vector Regression.

Q: What is the main advantage of using GPUs in Deep Learning compared to smaller datasets?

A:

QUESTION 15 Student Answer: ``python "EMPTY" `` Score: 0 Feedback: Empty answer. The student didn't provide any answer.